Jagonmoy Dey

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Research Interests

- Empirical Software Engineering • Applied AI/LLM for Software Engineering

Education

B.Sc. in Software Engineering *Jan 2018 – Feb 2023*
Shahjalal University of Science and Technology *Sylhet, Bangladesh*
CGPA: 3.60 / 4.0, Final 4 Semesters: 3.85+/4.00

Research Experience

Volunteer Researcher (Secure Software Engineering & AI-Generated Code) *November 2025 - March 2026*
Collaborating with Dr. Tanmay Bhowmik (Kennesaw State University) and his research group SERA (Software Engineering Research and Analysis Lab)
SE-AI-Security | *Python, Bash, CodeQL, Github API*

- Performing literature review on security weaknesses in AI-generated code and contributing to shaping a research proposal in Secure Software Engineering.
- Engineered a GitHub code-mining pipeline to collect AI-generated code datasets, handling pagination, rate limits, and file-size constraints, and exporting results to CSV.
- Automated local CodeQL scanning (cross-platform) to run query suites, extracting CWE-tagged security findings across codebases.
- Mapped CWEs to CVEs by scraping vulnerability intelligence (Python + Playwright), producing a consolidated dataset with CVSS, summaries, and publish/update dates.

Research Assistant (Thesis: Speaker Diarization with Speech Recognition in Bengali)
Shahjalal University of Science and Technology July 2022 - Feb 2023
View Thesis | *Python, TensorFlow, ELAN, Audacity*

- Contributed to a thesis on a multi-component speech processing system involving voice activity detection, speaker segmentation, embedding extraction, clustering, and speech recognition.
- Implemented the logic to integrate voice activity detection with speaker segmentation for accurate speaker turn detection, and integrated the Google Speech-to-Text API.
- Awarded **4.00/4.00** for the thesis and defense.

Work Experience

Software Engineer *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *July 2024 – Present*

- Built and scaled subrite.no, an Oslo-based multi-tenant SaaS subscription management platform, from scratch as a full-stack engineer, contributing to the core product and tenant projects while following Agile and SDLC practices.
- Architected email automation functions using Nest.js event queue system to handle both scheduled andevent-driven tasks, serving around 300,000 users.
- Optimized data fetching and caching with Next.js and Redis, boosting Google Lighthouse scores from 55%→95% (page load performance) and 85%→100% (SEO).
- Built a content pipeline integrating Sanity CMS with automated type generation and GROQ queries, featuring MJML email templates, rich text-to-HTML/PDF rendering, and optimized media delivery via MUX, Imgix, and Next/Image.
- Developing real-time control interfaces for OSC.no, an ocean simulation platform, processing high-frequency data streams via NATS pub/sub for remote equipment operation.
- Leveraged Claude Code with agentic workflows, hooks, skills, and MCP integrations to automate repetitive development tasks and streamline engineering pipelines.

Associate Software Engineer *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *Feb 2023 – June 2024*

- Integrated a tenant application with Subrite's OIDC system via NextAuth, handling secure authentication flows with dynamic redirect and callback URLs for tenant-specific login.
- Introduced Playwright for E2E testing and integrated it into the CI/CD pipeline, alongside Jest unit tests to ensure code reliability.
- Implemented dynamic tenant-specific custom fields, built responsive UIs, and developed reusable component libraries for a multi-tenant frontend.
- Optimized backend RESTful API performance and fine-tuned TypeORM SQL queries, resulting in a 1.5x improvement in API response times.

Trainee Software Engineer *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *Sep 2021 – February 2022*

- Received structured training on Web, HTTP, REST APIs, and software engineering foundations, and implemented these concepts by developing features for Cefalo's internal in-house web platforms.

Projects

AI-Driven Task Reviewer | *Python, FastAPI, ChromaDB* ▶ *View Demo*
Cefalo Bangladesh Limited

- Developed RAG + LLM workflows to detect duplicate tasks, incomplete specs, vague summaries, and unattended comments.
- Automated release notes and development summaries delivered via Jira, Slack, and email.

VocaBuilder - AI-Powered Vocabulary Builder | *Python, FastAPI, LangChain, ChromaDB, React*

- Built semantic retrieval and contextual story-based vocabulary learning powered by GPT-family LLMs.

Time-Series-with-stock | *Python, PyTorch, Scikit-learn, NumPy, Pandas*

- Extended TSLib with support for Kaggle stock datasets and unified model evaluation workflows.

SlackBook | *Javascript, Node.Js, Slack API, Facebook Graph API*

- Built a Slackbot enabling cross-platform message and attachment synchronization with Facebook.

Technical Skills

Languages: Python, JavaScript/TypeScript

Frameworks/Libraries: FastAPI, React.js, Next.js, Nest.js, Express.js, Playwright, Jest, PostgreSQL, Redis

ML/LLM Frameworks: LangChain, TensorFlow

LLM Technologies: ChromaDB, RAG Systems, LLM Integration (OpenAI, Gemini, etc.)

Developer Tools: Git, Docker, Claude, Postman

Competitive Programming

- Solved 1200+ problems; participated in 80+ contests; max **Codeforces** rating 1500+; Maintained multiple GitHub repositories for curated data structure and algorithmic resources, such as **Number Theory** and **Graph Theory**, which are used in the department as course materials.

Mentorship Experience

- Trainer for Junior Engineers, Cefalo — Led code reviews and workflow sessions.
- Competitive Programming Workshop Instructor, SUST — Conducted algorithm and problem-solving training.
- Contest Coordinator, SWE Technovent 2023 — Managed contest systems for 100+ universities.